Light tester

The light tester kit is designed to allow for the testing and repair of Incandescent Christmas light strands (this tester will not work with LED lamps). The kit includes a PCB and various components which, when assembled, create a device which can be used in testing the miniature incandescent bulbs along a strand to identify where burned out bulbs are located. Though testers like this can be purchased pre-assembled, the intention of this kit is to teach the concepts behind how the tester works so that those same concepts can be applied by the user toward future projects.

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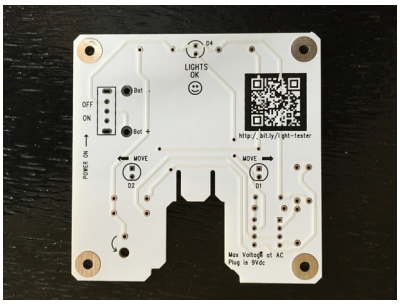
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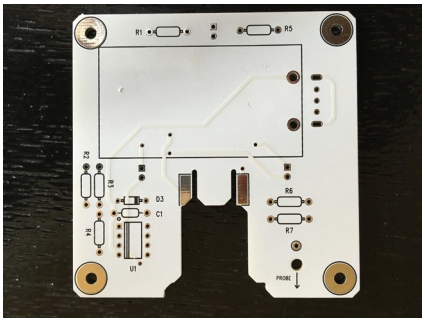
Photos

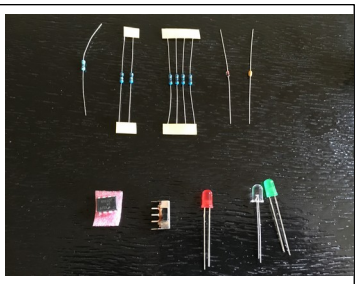


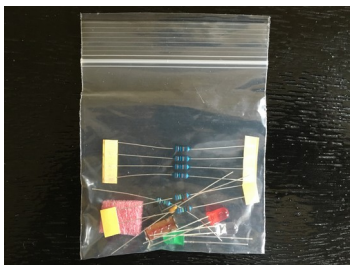

Wire for Making the Probe


Everything in the Box


PCB Top View


PCB Bottom View


All Components


Components in Bag


Standoffs and Hardware

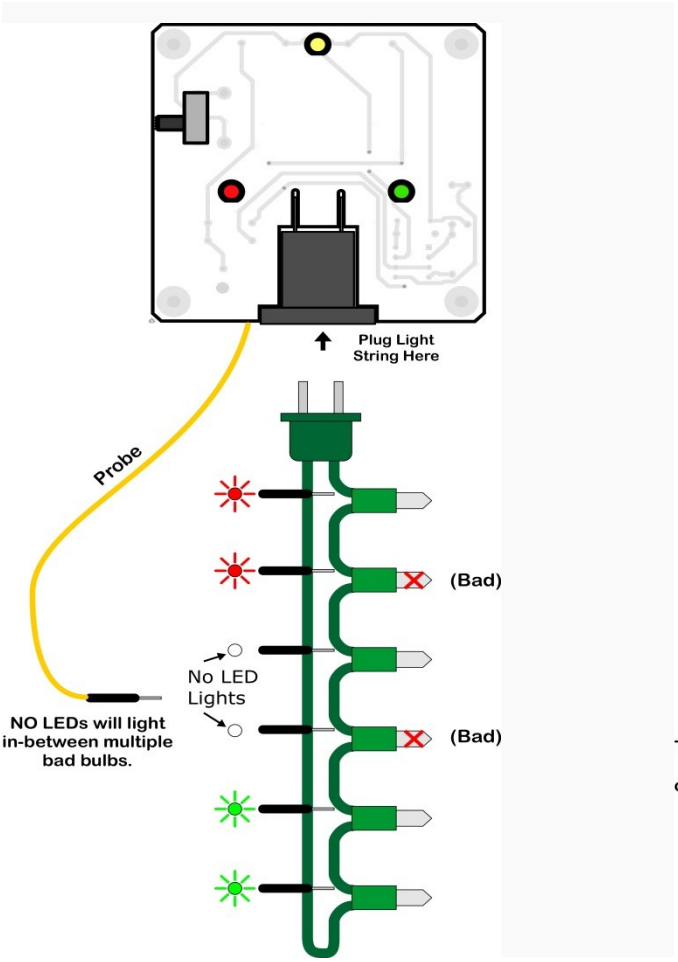
Instructions for Use

A strand of malfunctioning Christmas lights is plugged into the outlet on the PCB and several indicators guide the user in determining which bulb(s) need to be replaced in the strand. A probe is used to detect whether a specific bulb is working along the strand. The probe is used at the base of each bulb to make a physical electrical connection to the inside of the bulb contact.

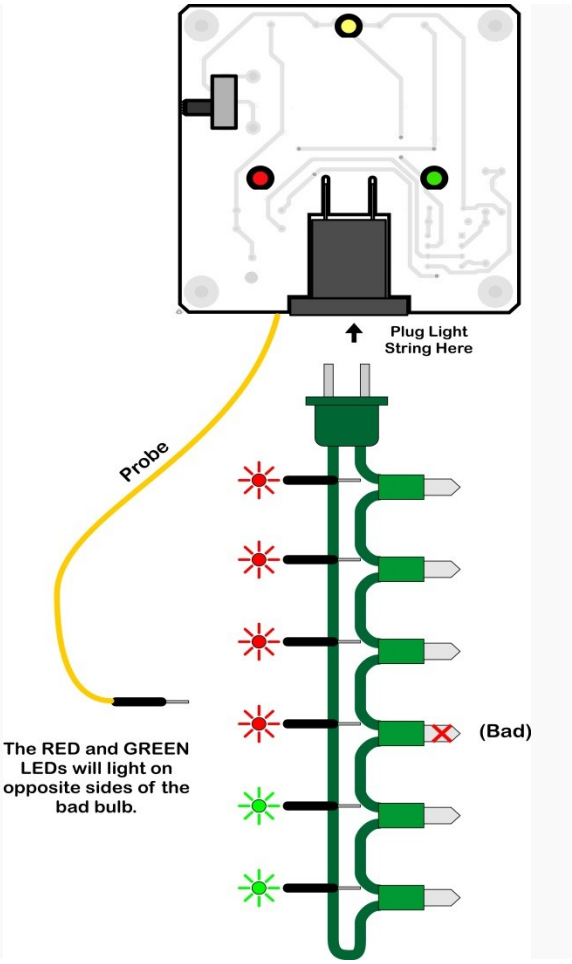
Safety: Disconnect the light string from household power before plugging it into the tester. The tester is powered only by its 9 V battery. Never connect the tester or a light string being tested to household power. Use only with miniature incandescent light strings; it does not work with LED strings.

There are 3 LEDs indicators on the light tester:

- Orange - Lights up with the entire strand is repaired.
- Green or Red – Indicates that at least one bulb is burned out.
- No LEDs lit – indicates that more than one bulb is bad.

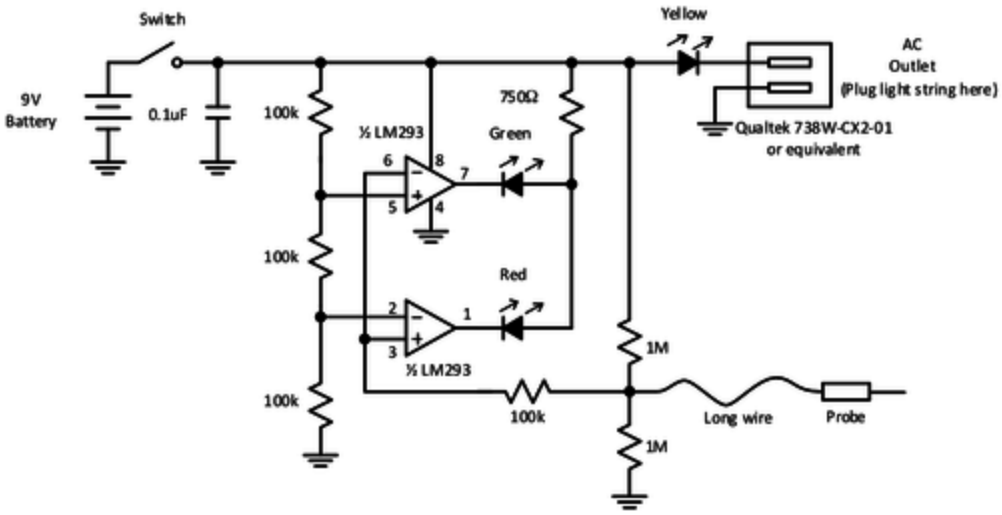


Example of a strand with 2 bad bulbs

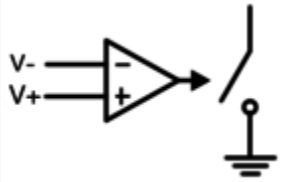


Example of a strand with 1 bad bulb

How it Works



LM293 Dual Comparator



If V+ is lower than V-, the switch is on. Otherwise the switch is off.

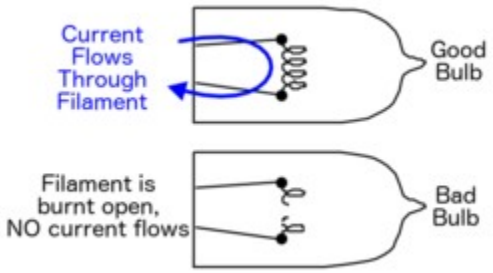
The LM293 is a dual comparator – a single package with two identical comparator circuits inside. The comparator compares the voltage on the two input pins (labeled “+” and “-“), then switches the output to a HIGH voltage or GROUND (0V) depending on the result. The LM293 output circuit acts like a switch. So when the output is “HIGH”, it leaves the switch OPEN. When the output is supposed to be LOW, it turns that switch ON, shorting the output to GROUND. If V+ is a higher voltage than V-, the output of the comparator is an open circuit. When V+ is lower than V-, the output is connected to GROUND.

In the schematic diagram of the tester, the inputs of the two comparators are biased using three 100k resistors. Those resistors divide the 9V battery voltage to 3V and 6V, which are applied to the inputs of the comparators. The other inputs of those comparators are connected to the flying probe, which is connected between two 100k resistors between 9V and ground. So the flying probe is normally at 4.5V when it’s not connected to anything.

Probe Not Touching

- Comparator 1 inputs are V+ = 6V, V- = 4.5V, output is OPEN, and the GREEN LED is OFF.
- Comparator 2 inputs are V+ = 4.5V, V- = 3V, output is OPEN, and the RED LED is OFF.

Light Strand Connected



When there is an open bulb, that bulb's filament is burned OPEN. The filament is a short piece of high-resistance wire, so think of a bad bulb as a wire that's been cut so that it's an open circuit. Since no current can flow in the circuit (there is no conductive path from the battery POSITIVE to the battery NEGATIVE through the light string), on one side of that bad bulb there is a conductive path to the BATTERY POSITIVE and on the other side of that bad bulb there is a path to the BATTERY NEGATIVE.

Probe Touching Positive Side

Now, when you use the probe and touch the POSITIVE side of this bulb, the PROBE is nearly 9V. Look back at the comparator circuit – with the probe nearly 9V, here is the state of the comparators:

- Comparator 1 inputs are V+ = 6V, V- = 9V, output is SHORTED to GROUND, and the GREEN LED is ON.
- Comparator 2 input are V+ = 9V, V- = 3V, output is still OPEN, and the RED LED is still OFF.

Probe Touching Negative Side

When you probe on the other side of the bad bulb, that side is connected to GROUND through the rest of the bulbs, so the comparators are like this:

- Comparator 1 inputs are V+ = 6V, V- = 0V, output is OPEN, and the GREEN LED is OFF.
- Comparator 2 inputs are V+ = 0V, V- = 3V, output is SHORTED to GROUND, and the RED LED is ON.

Special case: More than one broken bulb

What if more than one bulb is open? When two (or more) bulbs are open, there will still be the path to POSITIVE on one bad bulb, and there will be a path to GROUND from another of the bad bulbs. But in-between those two bulbs, there won't be a current path to GROUND or to POSITIVE. This isolated part of the circuit is not connected to anything. So if you probe in-between bad bulbs, the probe is still "floating", it won't be connected to POSITIVE or to GROUND, and so the probe voltage doesn't change, and neither of the LEDs lights. So if you probe on the string and nothing happens, this is the situation, and you need to probe toward the end (or beginning) of the string, until one of the LEDs lights up. Your first bad bulb will be the one with no indication on one side, and one of the LEDs lighting up on the other side. Replace that bad bulb, then continue probing as usual to find the other one(s) (could be several bad ones!).

Reverse Polarity Protection

The LM393 is the only component in the circuit that would be harmed by the battery being connected backwards. The 1N4148 diode blocks reverse voltage from the LM393, so that if the battery touches the battery contacts backwards, the circuit won't be damaged.

Included Parts

PCB Parts

Part	Quantity	Reference Number	Reference MPN
Cap, 0.1uF, 50V, 10%, X7R, Axial	1	C1	C410C104K5R5TA
Res, 1M ohm, 1%, 0.25W, 100ppm, Axial	2	R5, R6	MF1/4DC1004F
Res, 100K ohm, 1%, 0.25W, 100ppm, Axial	4	R2, R3, R4, R7	MF1/4DC1003F
Diode, Switching, 75V, 300mA, DO35 (THT)	1	D3	1N4148-T
Res, 750 ohm, 1%, 0.25W, 100ppm, Axial	1	R1	MF1/4DC750RF
Battery Holder, 9V, PCB Mount	1	M2	12BH611-GR
LED, Green, 30 deg, T1-3/4 (THT)	1	D1	LTL2R3KGD-EM
LED, Orange, 30 deg, T1-3/4 (THT)	1	D4	LTL2R3KFK
LED, Red, 30 deg, T1-3/4 (THT)	1	D2	LTL2R3KRD-EM
IC, Comparator, Dual, 5mV Vos, 32V Max, DIP8	1	U1	LM393AN
Sw, Slide, Rt Angle, 30V, 200mA, THT	1	M10	EG1206A

Hardware

Part	Quantity	Reference MPN
Screw, 4-40, PNH, SS, 0.25"	8	Bolt Depot 5312
Washer, flat, #4, medium, SS	4	Bolt Depot 5563
Outlet, AC, 2-prong	1	Qualtek 738W
Wire, 24AWG, Stranded (19x36l stranding), PVC	3ft. (0.91M)	Alpha Wire 1854/19 YL005
Heat shrink tubing, 1/8" ID, Black, 1" long	1in. (2.54cm)	
Battery, 9V, Alkaline	1	
Tape, double sided, 1/32" foam, 1" wide	2in (5.08cm)	3M 4462
Standoff, threaded, Aluminum, #4-40, Round, 0.75" Long	4	1688-440-AL

Reference MPN is the part used during the design process. Actual manufacturers may vary for parts included in the kit, though the parts will have the same specifications.

Part Identification

Resistors

Chart of Resistor Color Codes

Resistors

Value	1% Versions	5% Versions	Reference Designators
100kΩ	 BRN-BLK-BLK-ORG-BRN	 BRN-BLK-YEL-GOLD	R2, R3, R4, R7
1.0MΩ	 BRN-BLK-BLK-YEL-BRN	 BRN-BLK-GRN-GOLD	R5, R6
750 Ω	 YEL-GRN-BLK-BLK-BRN	 YEL-GRN-BRN-GOLD	R1

Example

100k 1% (BRN-BLK-BLK-ORG-BRN)



100k 5% (BRN-BLK-YEL-GOLD)



Other Components

0.1uF Capacitor

1N4148 Diode

LM393AN Integrated Circuit

Assembly

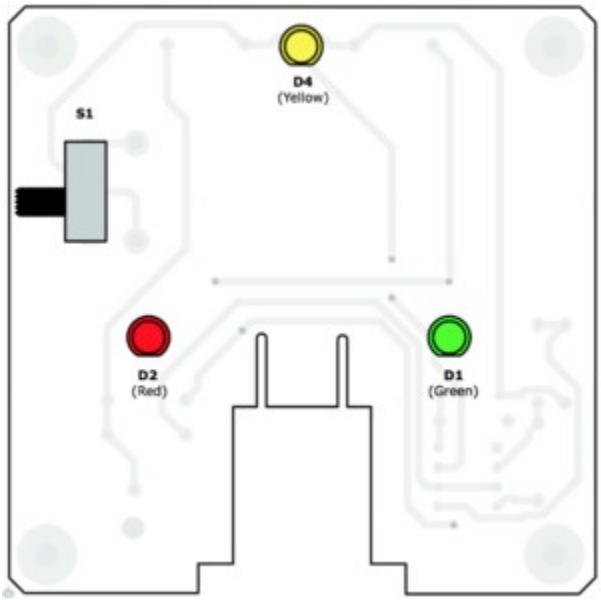
Assembly is meant to be done with very few tools:



Top Assembly

Solder the three LEDs and the switch.

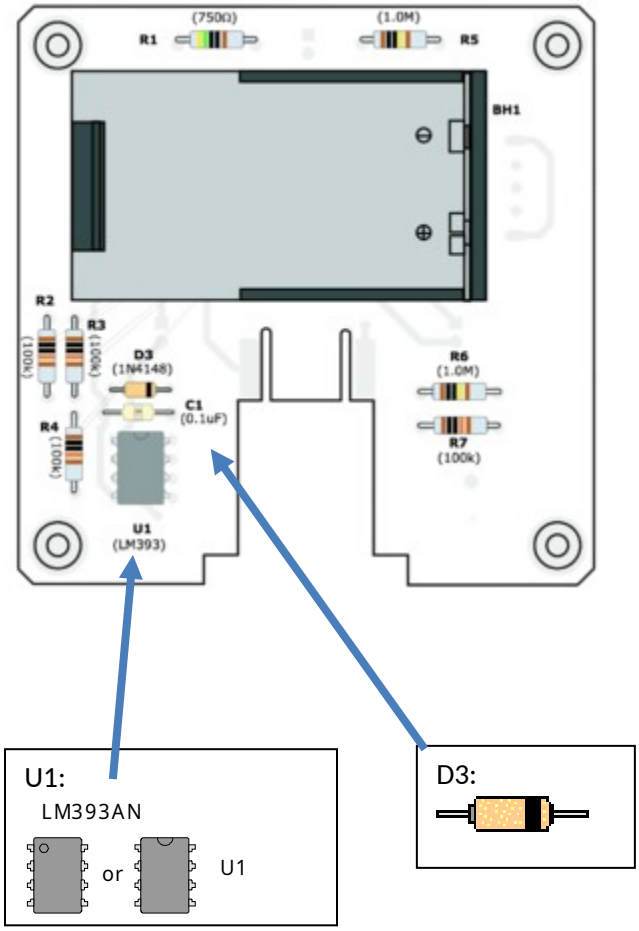
Note the polarity (flat side) of the LEDs!



Position	Part
D1	Green LED
D2	Red LED
D4	Yellow/Orange LED
S1	Power Switch

Bottom Assembly

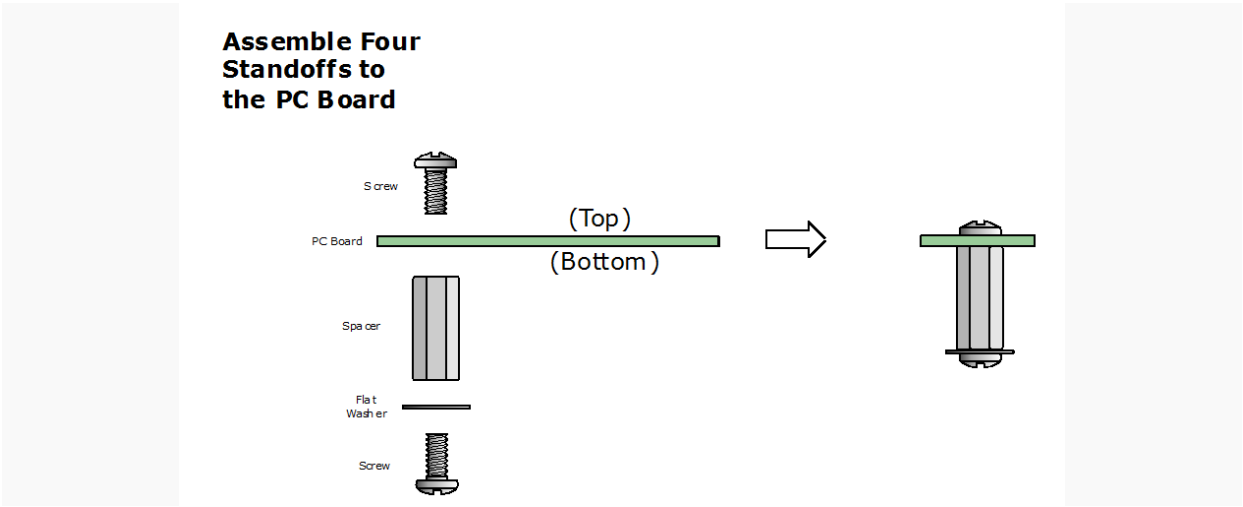
The rest of the parts get soldered to the bottom side of the board.
Resistors and Capacitors are not polarized – can be installed either way.
The diode D3 and the comparator U1 must be installed as shown.



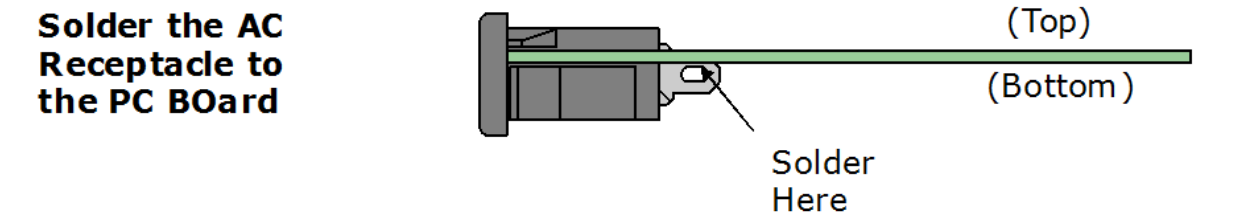
Position	Part
R2, R3, R4, R7	100K Resistor
R1	750 ohm Resistor
R5, R6	1M Resistor
C1	0.1uF Capacitor
U1	LM393 IC
BH1	Battery Holder
D3	1N4148 Diode

Final Assembly

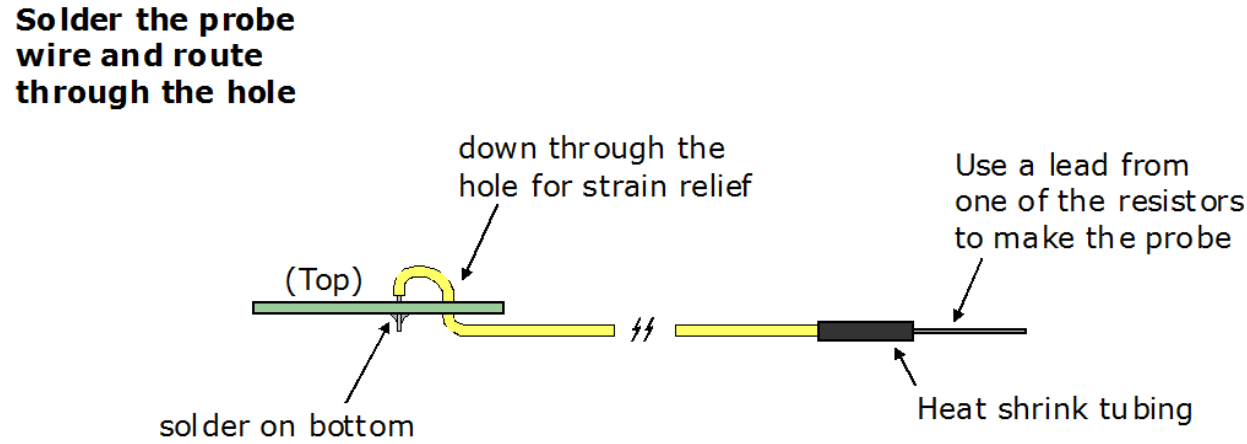
1. Assemble four standoffs to the PC board.
2. Solder the AC receptacle to the PC board.
3. Solder the Probe Wire and route through the hole.



Step 1 - Standoffs



Step 2 - Receptacle



Step 3 – Probe